

Highlights

Concentration Is Not Scaling: An Exact Decomposition of Aggregate Variance Elasticities

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- Concentration does not identify aggregate-volatility scaling.
- Variance elasticities admit an exact finite-interval decomposition.
- Signed covariance enters without logarithms of component values.
- Equal concentration paths can imply different diversification rates.
- Direct and reconstructed elasticities agree to machine precision.

Concentration Is Not Scaling: An Exact Decomposition of Aggregate Variance Elasticities

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ABSTRACT

Granular models relate aggregate volatility to the concentration of economic weights. This connection is exact only under restrictive assumptions on marginal shock variances, dependence, and population composition. Concentration is a state variable, whereas volatility scaling is an elasticity defined along a population path. We show that identical Herfindahl paths can generate different aggregate-variance elasticities and derive an exact finite-interval decomposition of that elasticity into changes in diagonal variance and signed covariance components. The decomposition uses the logarithmic mean of total variance and requires no logarithm of individual components. A contribution-weighted average of component elasticities emerges only under positivity and comparability conditions. Deterministic constructions verify the direct and reconstructed elasticities to machine precision. The result clarifies the boundary of concentration-based granular reasoning and supplies an accounting framework for micro-to-macro analyses of diversification.

1. Introduction

The granular approach restored an elementary but consequential fact to macroeconomic aggregation: units enter an aggregate according to their economic weights, so a small number of very large firms can prevent idiosyncratic fluctuations from averaging out rapidly (?). Under mutually independent shocks with a common variance, the variance of a weighted aggregate is proportional to the Herfindahl index

$$H(N) = \sum_{i=1}^N w_i(N)^2. \quad (1)$$

A fat-tailed firm-size distribution can therefore generate a concentration path that declines slowly as the population expands.

This benchmark is sometimes given a stronger interpretation: that the concentration path identifies the population-scaling rate of aggregate volatility. That conclusion is not general. Concentration describes the distribution of weights at a given scale. Aggregate-volatility scaling is an elasticity describing how the complete variance–covariance structure changes along a specified population path. The two objects coincide only when restrictions on marginal variances, dependence, and composition supply the missing bridge between them.

For the weighted aggregate

$$X_t(N) = \sum_{i=1}^N w_i(N) \varepsilon_{i,t}(N), \quad (2)$$

aggregate variance is

$$V(N) = \sum_{i=1}^N w_i(N)^2 \text{Var}[\varepsilon_{i,t}(N)] + 2 \sum_{i < j} w_i(N) w_j(N) \text{Cov}[\varepsilon_{i,t}(N), \varepsilon_{j,t}(N)]. \quad (3)$$

The proportionality $V(N) = \sigma^2 H(N)$ follows when all shocks share a variance σ^2 that is stable along the population path and the off-diagonal terms vanish. These are not incidental simplifications. Even under independence, allowing the common shock variance to vary with population scale gives

$$-\frac{d \log V(N)}{d \log N} = -\frac{d \log H(N)}{d \log N} - \frac{d \log \sigma^2(N)}{d \log N}. \quad (4)$$

The concentration elasticity identifies the variance elasticity only when the second term is zero. Heterogeneous firm-level variances replace $H(N)$ by a variance-weighted concentration object, while common shocks and economic linkages add off-diagonal terms. Population expansion may also alter the size, sectoral, or network composition of the aggregate. Once these channels are present, the Herfindahl index is not a sufficient statistic for either aggregate variance or its population elasticity.

This paper separates the granular insight from that stronger inference. Granularity correctly establishes that aggregate variance is weighted by firm shares and may be dominated by a small number of large units. It does not, by itself, establish that the firm-size distribution identifies the rate at which aggregate variance changes with population size. We make this distinction precise in two steps.

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First, we show that identical concentration paths can generate different aggregate-volatility scaling. Equal-weight populations all satisfy $H(N) = 1/N$, but aggregate variance follows $1/N$ only under independence. With persistent positive correlation, it instead approaches a positive covariance floor and its scaling elasticity converges to zero. No statistic constructed solely from the weights can distinguish the two systems.

Second, we derive an exact finite-interval decomposition of the aggregate variance elasticity into changes in diagonal variance and signed covariance components. The formula does not take logarithms of individual components and therefore remains well defined when covariance contributions are negative. It also identifies the restrictive positive-component case in which an aggregate elasticity may be read as a contribution-weighted average of component elasticities.

The paper sits at the intersection of three established literatures. Granular models study how unequal weights transmit microeconomic shocks to aggregates; network models show that dependence and propagation can prevent diversification; and index-decomposition and risk-allocation methods provide exact additive accounting tools. Our claim is deliberately narrower than invention of any of these ingredients. The contribution is their combination for a distinct micro-to-macro object: the finite-interval elasticity of aggregate variance with respect to population scale. This formulation separates a level statement about weighted exposure from a path statement about diversification, retains signed covariance contributions, and yields an identity that can be checked directly in empirical or synthetic data.

2. Exact decomposition

Consider an aggregate composed of Q contribution series,

$$X_t(N) = \sum_{q=1}^Q c_{q,t}(N), \quad (5)$$

where $N > 0$ indexes population scale. Its variance is

$$V(N) = \sum_{q=1}^Q \text{Var}[c_{q,t}(N)] + 2 \sum_{q < r} \text{Cov}[c_{q,t}(N), c_{r,t}(N)]. \quad (6)$$

Let $\mathcal{K}$ collect the diagonal terms $C_{qq}(N) = \text{Var}[c_{q,t}(N)]$ and the doubled off-diagonal terms $C_{qr}(N) = 2 \text{Cov}[c_{q,t}(N), c_{r,t}(N)]$ for $q < r$. Then

$$V(N) = \sum_{k \in \mathcal{K}} C_k(N). \quad (7)$$

Diagonal components are non-negative, whereas covariance components may have either sign. The component

universe $\mathcal{K}$ is assumed to retain a comparable meaning at the population levels being contrasted.

For $N_0, N_1 > 0$, $N_0 \neq N_1$, and positive aggregate variance at both levels, define the finite-interval variance elasticity

$$\hat{\alpha}(N_0, N_1) = -\frac{\log V(N_1) - \log V(N_0)}{\log N_1 - \log N_0}. \quad (8)$$

For $a, b > 0$, let

$$L(a, b) = \begin{cases} \frac{a - b}{\log a - \log b}, & a \neq b, \\ a, & a = b, \end{cases} \quad (9)$$

be the logarithmic mean.

Proposition 1 (Exact finite-interval decomposition). *For each $k \in \mathcal{K}$, define*

$$\hat{\alpha}_k^C(N_0, N_1) = -\frac{C_k(N_1) - C_k(N_0)}{L(V(N_1), V(N_0)) [\log N_1 - \log N_0]}. \quad (10)$$

Then

$$\hat{\alpha}(N_0, N_1) = \sum_{k \in \mathcal{K}} \hat{\alpha}_k^C(N_0, N_1). \quad (11)$$

The identity remains valid when individual covariance components are negative.

Proof. Equation (??) implies

$$\sum_{k \in \mathcal{K}} [C_k(N_1) - C_k(N_0)] = V(N_1) - V(N_0). \quad (12)$$

Summing equation (??) over k and using

$$\frac{V(N_1) - V(N_0)}{L(V(N_1), V(N_0))} = \log V(N_1) - \log V(N_0) \quad (13)$$

produces equation (??) exactly. $\square$

The decomposition attributes the aggregate elasticity to changes in the underlying variance and covariance components without requiring their individual logarithms. Signed covariance terms therefore enter directly, provided total variance remains positive. The result is exact for each finite interval and does not require a global power law. A regression coefficient estimated over many population levels may still provide a useful summary, but it averages interval-specific elasticities whose component weights can change across the fitted range. The decomposition therefore distinguishes an accounting identity from a parametric claim: it reconstructs what drives the observed change in aggregate variance without assuming that one invariant exponent governs every scale. In the differentiable limit,

$$\alpha_{\text{eff}}(N) = -\frac{d \log V(N)}{d \log N} = -\sum_{k \in \mathcal{K}} \frac{N C'_k(N)}{V(N)}. \quad (14)$$

If every component is positive, defining $\alpha_k(N) = -d \log C_k(N) / d \log N$ gives

$$\alpha_{\text{eff}}(N) = \sum_{k \in \mathcal{K}} \frac{C_k(N)}{V(N)} \alpha_k(N). \quad (15)$$

This resembles a weighted average, but its weights generally vary with N . A fixed average of local exponents arises only under additional restrictions, such as common component elasticities or contribution shares that remain invariant along the population path. When positive components decay at different power-law rates, their shares change with N and the slowest-decaying component eventually dominates. With signed covariance components the normalized contributions need not be non-negative, so the representation is not a convex combination and the aggregate elasticity need not lie within the range of the diagonal elasticities. Covariance is therefore not merely a correction to the level of aggregate variance; its own scale dependence can determine the limiting diversification rate.

Comparability across scales is substantive. If parts appear, disappear, or change meaning between N_0 and N_1 , the measured elasticity may combine diversification with composition, partition, or dependence changes. The identity remains valid only after those changes are represented through a common accounting structure.

3. Why concentration does not identify volatility scaling

Consider the weighted aggregate

$$X_t(N) = \sum_{i=1}^N w_i(N) \varepsilon_{i,t}(N), \quad \sum_{i=1}^N w_i(N) = 1. \quad (16)$$

For equal weights, $w_i(N) = 1/N$, the concentration path is

$$H(N) = \frac{1}{N}. \quad (17)$$

Suppose first that shocks have unit variance and are mutually independent. Aggregate variance is then

$$V_{\text{iid}}(N) = \text{Var} \left[\frac{1}{N} \sum_{i=1}^N \varepsilon_{i,t} \right] = \frac{1}{N}, \quad (18)$$

and its local variance elasticity is

$$\alpha_{\text{iid}}(N) = 1. \quad (19)$$

Now retain the same weights and marginal variances but impose a common pairwise correlation $\rho \in (0, 1)$. Aggregate variance becomes

$$\begin{aligned} V_\rho(N) &= \frac{1}{N^2} [N + \rho N(N-1)] \\ &= \rho + \frac{1-\rho}{N}, \end{aligned} \quad (20)$$

figures/figure_1_concentration_is_not_scaling.pdf

Figure 1: Equal concentration does not imply equal volatility scaling. Both populations have concentration $H(N) = 1/N$. Aggregate variance declines as $1/N$ under independent shocks but approaches a positive floor when shocks share a common correlation component.

with local elasticity

$$\alpha_\rho(N) = \frac{1-\rho}{\rho N + 1 - \rho}. \quad (21)$$

The independent aggregate continues to diversify at rate N^{-1} , whereas the correlated aggregate approaches the variance floor $\lim_{N \rightarrow \infty} V_\rho(N) = \rho$ and $\lim_{N \rightarrow \infty} \alpha_\rho(N) = 0$.

The two economies have the same number of units, weights, Herfindahl index, effective number of firms, and entire concentration path. Yet they have different variance paths and limiting elasticities. The missing information is not a refinement of concentration: it is the scale-dependent variance-covariance structure of the contributions. This distinction also rules out a common empirical shortcut. Estimating a power law for $H(N)$ and interpreting its slope as the population exponent of aggregate variance is valid only after the micro-variance and covariance terms have been shown to be scale neutral. Concentration and aggregate-volatility scaling are therefore different estimands even when they coincide under the independent homogeneous benchmark.

This result qualifies rather than overturns the granular mechanism. The presence of large firms may still make weighted idiosyncratic shocks quantitatively important. What does not follow is that the population elasticity of aggregate variance can be recovered from the firm-size distribution alone. The Herfindahl connection is conditional on stable homogeneous shock variances, independence, and a coherent population path. Relaxing those conditions does not merely add noise around the granular prediction; it changes the estimand that concentration can identify.

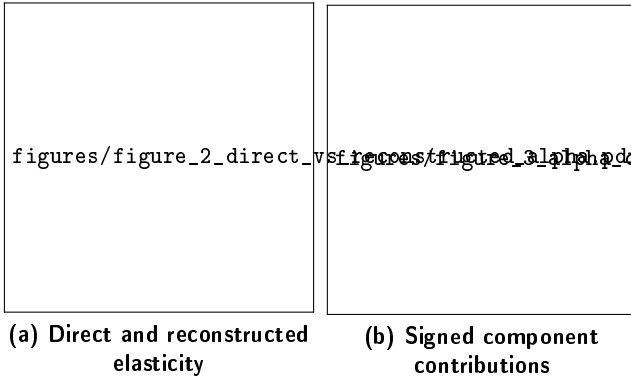


Figure 2: Exact reconstruction of aggregate variance scaling. Panel (a) compares the directly measured finite-interval elasticity with its reconstruction. Panel (b) separates diagonal and signed covariance contributions. Differences are at floating-point precision.

4. Deterministic computational verification

We verify the variance and scaling identities using deterministic finite-sample constructions rather than Monte Carlo approximations. For $N \in \{4, 8, 16, 32, 64\}$, we construct contribution panels with $T = 256$ observations and fixed seed 20260723. Centered random matrices are orthogonalized by QR decomposition and scaled to have identity sample covariance under $ddof = 1$. Cholesky transformations then impose known target covariance matrices, making measured and designed moments agree up to floating-point precision.

The first construction reproduces the independent and correlated systems in Figure ???. Two additional constructions contain heterogeneous diagonal components, first without and then with a signed covariance contribution. At each N , aggregate variance is measured directly from the summed contribution series and reconstructed from diagonal variances and doubled covariance terms. The finite-interval elasticity is then computed directly from adjacent aggregate variances and reconstructed through Proposition ??.

The maximum absolute residual is 4.441×10^{-16} for the variance identity and 1.221×10^{-15} for the scaling identity; the largest discrepancy between designed and measured covariance components is 5.551×10^{-16} . These calculations verify the implementation and numerical consistency of the accounting chain. They do not constitute empirical evidence about the scaling behavior of any particular economy.

5. Conclusion

Concentration locates aggregate exposure; it does not generally identify how aggregate variance changes as the population expands. The Herfindahl connection requires a stable homogeneous variance structure, restricted covariance, and a coherent population path.

We provide an exact finite-interval decomposition that attributes the aggregate variance elasticity to changes in diagonal and signed covariance components without logging the components themselves. The result preserves the central granular insight while defining its boundary: the firm-size distribution can make large firms important, but it is not a sufficient statistic for population scaling. The decomposition supplies a transparent accounting contract for empirical micro-to-macro work, shifting the empirical task from reading an exponent off the size distribution to measuring how variance and covariance contributions change along an explicit population path.

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Data and code availability

No empirical data were used in this study. Source code, deterministic computational constructions, tests, and generated verification artifacts will be made available in a public repository.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used OpenAI ChatGPT and Codex to assist with research organization, software implementation, adversarial checking, and language revision. The author reviewed and verified the mathematical derivations, computational outputs, references, and final text, edited the resulting material as needed, and takes full responsibility for the content of the publication.